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On the two-year anniversary of the Paris climate accord, the world's government, civic and business leaders are coming together in Paris to discuss one of the most important issues and opportunities of our time, climate change. I'm excited to lead the Microsoft delegation at these meetings. While the experts' warnings are dire, at Microsoft we believe technology advances can help us better understand and address the environmental issues facing our planet. That's why we're announcing in Paris that we are broadening our AI for Earth program with an expanded strategic plan and committing $50 million over the next five years to put artificial intelligence technology in the hands of individuals and organizations around the world who are working to protect our planet.

At Microsoft, we believe artificial intelligence is a game changer. Our approach as a company is focused on democratizing AI so its features and capabilities can be put to use by individuals and organizations around the world to improve real-world outcomes. There are few societal areas where AI can be more impactful than in helping address the urgent work needed to monitor, model and manage the earth's natural systems.

Data can help tell us about the health of our planet, including the conditions of our air, water, land and the well-being of our wildlife. But we need technology's help to capture this vast amount of data and convert it into actionable intelligence. AI can be trained to classify raw data from sensors on the ground, in the sky or in space into categories that both humans and computers understand. Fundamentally, AI can accelerate our ability to observe environmental systems and how they are changing at a global scale, convert the data into useful information and apply that information to take concrete steps to better manage our natural resources.

Already, we are seeing the transformative potential of AI. In the energy sector, companies like [Agder Energi], a utility in Norway that produces renewable energy, are using Microsoft's cloud and AI to better capture, analyze and act on the intelligence gathered across its electrical grid. Through these technologies, Agder is now able to predict and prepare for vacillating energy needs in response to changes in demand as electric vehicles increasingly tax Norway's grid; data and AI have enhanced the performance of existing infrastructure, reducing the need for expensive new projects. AI is helping create a more effective, reliable and autonomous grid, while enabling customers and the country to consume more renewable energy as it transitions to a more electricity-based future.

In a similar way, we're seeing new AI and cloud technologies being used to improve the electrical efficiency of buildings. In Singapore, [JTC], responsible for the development of the nation's industrial infrastructure, has centralized its operations on the Microsoft Cloud to monitor, analyze and optimize 39 of its buildings. Using sensor data and analytics, JTC can now identity and rectify faults before breakdowns occur, resulting in a 15 percent drop in energy cost avoidance in the first three buildings.

It's worth imagining what these types of steps can mean if we can help bring them to scale globally. Estimates suggest that in the United States, buildings are responsible for about 40 percent of total energy consumption. That means an efficiency improvement of even 15 percent in buildings globally would translate into a 6 percent reduction in global energy consumption. And as AI continues to advance, we have the opportunity to learn more and aim even higher.

We're seeing a similar cause for optimism when we see how AI is being used in agriculture. In Australia, high labor and import costs, dry weather and the highest variability in climate of any country in the world make farming increasingly challenging. The Yield, a Tasmanian ag-tech company, has created a solution that uses sensors, analytics and apps to produce real-time weather data, right down to field level, helping growers make smarter decisions that can reduce their use of water and other inputs while also increasing their yield. In the sea, the Yield is working with local oyster farmers to create the first product to increase aquaculture production using machine learning. The solution already has reduced harvest closures caused by rain by 30 percent, giving growers back four weeks a year of harvest time.

It's these kind of results that motivated us to step up our ambition when it comes to AI for Earth. As we look to the future, we're committed to working with farmers around the world. We envision a future with broadband connectivity for every farm and internet sensors for every acre of land. Building on cutting edge-work in Microsoft Research, we'll help farmers put AI to work not only to better analyze soil and rainfall conditions, but also to use predictive analytics to improve agricultural yields and reduce adverse environmental impacts. With the world's population continuing to grow, these changes cannot come fast enough.

At Microsoft, we believe AI for Earth will be a force multiplier for groups and individuals like these who are creating sustainable solutions. That's why we're not just putting more resources into this effort, but also coupling this with a long-term commitment to applying AI to grow and scale in four key areas – climate, water, agriculture and biodiversity.

We'll do this in three ways. First, we'll expand seed grants around the world to create and test new AI applications. Since our launch of AI for Earth six months ago, Microsoft has awarded over 35 grants in more than 10 countries for access to Microsoft Azure and AI technology. We will also provide universities, nongovernmental organizations and others with advanced training to put AI to its best use. Already, we're seeing success around the world in projects that are putting AI to work on climate, water, agriculture and biodiversity.

Next, as these projects and our work in this area matures, we will identify the projects that show the most promise and make larger investments to help bring them to scale. We'll do this not only by providing greater resources for these projects, but also by partnering closely and working in depth with a new multi-disciplinary team at Microsoft that will bring together AI and sustainability subject matter experts. As we help groups scale promising AI technology solutions, we'll help them commercialize these services, so they can have a global impact as quickly and broadly as possible. These will be in addition to our existing efforts: enabling real-time precision conservation and improving land cover mapping, precision agriculture to increase yield with fewer resources with FarmBeats and more efficient, effective biodiversity tracking and protection approaches through Project Premonition.

Finally, as these projects advance, we'll identify and pursue opportunities to incorporate new AI advances into platform-level services so that others can use them for their own sustainability initiatives. Some of this will involve platform services that will be offered by others. In other instances, these may be incorporated into Microsoft's own platform services.

We face a collective need for urgent action to address global climate issues. When we think about the environmental issues we face today, science tells us that many are the product of previous Industrial Revolutions. As we enter the world's Fourth Industrial Revolution, a technology-fueled transformation, we must not only move technology forward, but also use this era's technology to clean up the past and create a better future.

Tags: artificial intelligence, Environment